

Nested Queries in DBMS (Subqueries)

1. Meaning of Nested Query

A nested query is a query written inside another query.

The inner query gives output to the outer query.

It is also called a subquery.

Basic structure:

```
SELECT column_name
```

```
FROM table
```

```
WHERE column_name = (SELECT column_name FROM table WHERE  
condition);
```

2. Types of Nested Queries

1. Single Row Subquery

This returns only one value (one row).

Used with operators like =, >, <, >=, <=

Example:

Find employee with highest salary

```
SELECT name
```

```
FROM employee
```

```
WHERE salary = (SELECT MAX(salary) FROM employee);
```

Inner query finds maximum salary

Outer query finds employee with that salary

2. Multiple Row Subquery

This returns more than one value.

Used with IN, ANY, ALL

Example:

```
SELECT name  
FROM employee  
WHERE dept_id IN (SELECT dept_id FROM department WHERE  
dept_id IN (10,20));
```

3. Subquery using IN

Used when multiple values are returned

Example:

Find students who scored more than 80

```
SELECT name  
FROM student  
WHERE id IN (SELECT student_id FROM marks WHERE score > 80);
```

4. Subquery using EXISTS

Checks whether inner query returns any result

Example:

```
SELECT name  
FROM student s  
WHERE EXISTS (SELECT * FROM marks m WHERE m.student_id =  
s.id);
```

Returns students who have marks

5. Subquery using NOT EXISTS

Opposite of EXISTS

Example:

```
SELECT name  
FROM student s  
WHERE NOT EXISTS (SELECT * FROM marks m WHERE m.student_id =  
s.id);
```

Returns students without marks

6. Correlated Subquery

Inner query depends on outer query.
It runs once for each row.

Example:

```
SELECT name  
FROM employee e1  
WHERE salary > (SELECT AVG(salary)  
FROM employee e2  
WHERE e1.dept_id = e2.dept_id);
```

Employees earning more than department average

7. Subquery using ALL

Compares value with all results

Example:

```
SELECT name  
FROM employee  
WHERE salary > ALL (SELECT salary FROM employee WHERE dept_id  
= 10);
```

Salary greater than all employees in dept 10

8. Subquery using ANY

Compares value with any one result

Example:

```
SELECT name  
FROM employee  
WHERE salary > ANY (SELECT salary FROM employee WHERE dept_id  
= 10);
```

Salary greater than at least one employee in dept 10

3. Important Points

- Subquery is always written inside parentheses
- Inner query executes first
- Outer query uses inner query result
- Can be used in SELECT, FROM, WHERE clauses
- Helps in solving complex queries easily

4. Advantages

- Breaks complex queries into simple parts
- Easy to understand logic
- Improves query flexibility
- Useful for filtering data

5. Disadvantages

- Can be slower in execution
- Hard for beginners in some cases
- Debugging is difficult

6. Difference Between Subquery and Join

Subquery:

- Query inside another query
- Easy to understand
- Used for filtering

Join:

- Combines multiple tables
- Faster in many cases
- Used for merging data

7. Simple Example

Find students whose marks are above average

SELECT name

FROM student

WHERE marks > (SELECT AVG(marks) FROM student);

8. Definition (for exams)

A nested query is a query inside another query where the result of the inner query is used by the outer query.